

AHSAN SHAHZAD

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI/ML Engineer with 1+ year of hands-on experience building production GenAI systems, including RAG and multi-agent LangGraph workflows. Solid foundation in ML/DL, with practical experience in LLM evaluation, observability, and experiment tracking.

EDUCATION

B.S. in Software Engineering Comsats University Islamabad, Lahore Campus.	Sep 2021 – June 2025
Fsc Pre Engineering Government College University, Lahore.	Sep 2019 – June 2021

WORK EXPERIENCE

Associate Software Engineer / Folium AI June 2025 – present

Python | FastAPI | LangChain | LangGraph | Pinecone | LLMs | Docker | Git | LangSmith | AWS

- Built an **agentic RAG chatbot** for a crypto education platform, with multi-format ingestion, Pinecone retrieval and web search. Reached **95%+ accuracy** on manual review and about **3s time to first token**, with LangSmith tracing for p50/p95 latency and cost.
- Architected a **multi-agent LangGraph** system on FastAPI that turns messy medical-billing documents into validated decisions and letters, using vision-based classification, parallel pipelines, conditional routing and **Pydantic structured outputs**. Built its document layer on **AWS Textract and S3**, where async jobs rebuild tables and forms into LLM-ready markdown and **quality gates** combine LLM and rule-based checks to flag low-confidence pages.
- Developed a youth-savings assistant: a multi-intent router feeding a ReAct tool-calling agent that queries live user data through Django ORM tools, with multimodal input (images, docs, voice), guardrails, and model tiering that uses a lighter model for tool selection and a stronger one for responses.
- Built TOC and blog-generation pipelines for an SEO content platform, working directly with the SEO team to gather requirements and iterate on feedback. Shipped **SERP-driven competitor analysis, claim verification** and **Claude/OpenAI fallback**.
- Containerised services with Docker and set up GitHub Actions CI with pytest regression tests ruff/mypy and pre-commit. Mentored an intern on Git and production workflows.

PROJECTS

Chest Cancer Classification: End-to-End MLOps Keras · DVC · MLflow · FastAPI · Docker

- Config-driven, modular pipeline that classifies CT scans by **fine-tuning VGG16**. It's orchestrated as a **DVC pipeline**, so only stages affected by a change re-run.
- Experiments and versioned models tracked in the **MLflow registry**, with the model served through a **containerised FastAPI** inference API.

System Design Mentor: Hybrid RAG with Evals LangGraph · Pinecone · DeepEval · LangSmith

- Built a **hybrid-search RAG** assistant over five system design and ML books, combining dense and keyword (**BM25**) retrieval with a **reranker** and metadata-aware chunking.
- Eval suite** with golden datasets for retrieval, answer quality and latency. **Regression checks** compare each change against a baseline.

SKILLS

Languages:	Python, TypeScript, JavaScript (ES6+), SQL, C/C++, Java
Frameworks & Libraries:	FastAPI, Django, Pydantic, React.js, Next.js, Node.js, Tailwind CSS
AI & ML Tools:	LangChain, LangGraph, DeepEval, MCP, PyTorch, TensorFlow, Scikit-learn, Hugging Face
Databases:	MongoDB, PostgreSQL, Supabase, Pinecone, chroma, Neo4j
Developer Tools:	AWS, Docker, Git, GitHub Actions (CI/CD), LangSmith, Jupyter, Jira, MLFlow, DVC
Core Skills:	LLM Evaluation, RESTful APIs, Asynchronous Programming, OOP, DSA, Agile

CERTIFICATIONS

LangSmith Essentials | **React Basics** (HackerRank) | **NodeJS Intermediate** (HackerRank)